Introduction to Descriptive Statistics

1. What is Descriptive Statistics?

Descriptive statistics is the branch of statistics that summarizes and describes the main features of a dataset. Instead of drawing conclusions about a larger population, it simply organizes data so that it is easy to understand. Descriptive statistics uses numbers, tables, and graphs to show what the data looks like. The three main types are measures of central tendency, measures of dispersion, and measures of shape.

2. Measures of Central Tendency

Measures of central tendency describe the center of a dataset with a single value. The mean is the sum of all values divided by the number of values. For example, the mean of 2, 4, 6, 8 is 5. The median is the middle value when the data is sorted. If there is an even number of values, the median is the average of the two middle values. The mode is the value that appears most often. A dataset can have one mode, more than one mode, or no mode at all.

The mean is sensitive to outliers, which are extremely high or low values. The median is more robust and is preferred when the data is skewed, such as income or house prices.

3. Measures of Dispersion

Measures of dispersion describe how spread out the data is. The range is the difference between the maximum and minimum values. The variance is the average of the squared differences from the mean. The standard deviation is the square root of the variance, and it is expressed in the same units as the data. A small standard deviation means the values are close to the mean, while a large standard deviation means the values are widely spread out.

The interquartile range (IQR) is the difference between the third quartile (Q3) and the first quartile (Q1). It describes the spread of the middle 50 percent of the data and is not affected by outliers.

4. Measures of Shape

Skewness measures the asymmetry of a distribution. A positively skewed distribution has a long tail on the right, and a negatively skewed distribution has a long tail on the left. A symmetrical distribution, like the normal distribution, has zero skewness. Kurtosis measures how heavy the tails of a distribution are compared to a normal distribution.

5. Data Visualization

Graphs help us understand data quickly. A histogram shows the frequency distribution of continuous data. A bar chart compares categories. A box plot displays the median, quartiles, and outliers of a dataset. A pie chart shows proportions of a whole. A scatter plot shows the relationship between two variables.

6. Types of Data

Data can be qualitative or quantitative. Qualitative (categorical) data describes qualities, such as color or gender. It can be nominal, where categories have no order, or ordinal, where categories have a meaningful order such as low, medium, and high. Quantitative (numerical) data represents measurable quantities. It can be discrete, such as the number of students in a class, or continuous, such as height or weight.

7. Example: Class Test Scores

Suppose a class of five students scored 60, 70, 70, 80, and 100 on a test. The mean score is 76. The median score is 70. The mode is 70. The range is 40 because the highest score is 100 and the lowest is 60. The score of 100 pulls the mean above the median, so this dataset is slightly positively skewed.

8. Why Descriptive Statistics Matters

Descriptive statistics is the first step in any data analysis. It helps researchers check data quality, find errors and outliers, and understand patterns before using inferential statistics. Teachers, businesses, doctors, and scientists all use descriptive statistics to summarize results and make decisions.